

LECTURE 6

Volatility Modelling: ARCH, GARCH and Multivariate Extensions

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Agenda

- 01 Stylised facts of financial returns
- 02 ARCH model (Engle, 1982)
- 03 GARCH model (Bollerslev, 1986)
- 04 Asymmetric GARCH: EGARCH and GJR
- 05 Estimation and diagnostic testing
- 06 Multivariate GARCH: DCC
- 07 Value-at-Risk and Expected Shortfall
- 08 Portfolio and risk management applications

Daily USD/BRL exchange rate returns and IBOVESPA returns (2000–2023). We use this series throughout to illustrate every model and test.

Section 1

Stylised Facts of Financial Returns

What Makes Financial Returns Special? I

Let $r_t = \ln(P_t/P_{t-1})$ be the log-return of an asset price.

Stylised facts (universally observed)

- 1 **Near-zero autocorrelation** of r_t (market efficiency)
- 2 **Volatility clustering**: large changes tend to be followed by large changes; small by small
- 3 **Fat tails** (leptokurtosis): extreme returns occur far more often than the normal distribution predicts
- 4 **Leverage effect**: negative returns increase volatility more than positive returns of equal magnitude
- 5 **Long memory in volatility**: $|r_t|$ or r_t^2 have slowly decaying ACF

What Makes Financial Returns Special? II

If volatility clusters, the errors ε_t in an OLS regression are:

- Conditionally heteroskedastic
- Uncorrelated but **not** independent
- Excess kurtosis $\Rightarrow$ standard errors are wrong
- VaR based on normality is severely underestimated in turbulent periods

ARCH/GARCH models explicitly capture these features.

What Makes Financial Returns Special? III

Testing for ARCH effects

Engle (1982) LM test: Regress $\hat{\varepsilon}_t^2$ on q lags:

$$\hat{\varepsilon}_t^2 = \omega + \alpha_1 \hat{\varepsilon}_{t-1}^2 + \cdots + \alpha_q \hat{\varepsilon}_{t-q}^2 + \nu_t$$

Test $H_0 : \alpha_1 = \cdots = \alpha_q = 0$ using $TR^2 \sim \chi^2(q)$ under H_0 .

- 1 Fit ARMA model to r_t , obtain $\hat{\varepsilon}_t$
- 2 Plot $\hat{\varepsilon}_t^2$ — visible clustering?
- 3 ACF of $\hat{\varepsilon}_t^2$ — significant spikes?
- 4 Apply Engle LM test ($q = 5$ or $q = 10$)
- 5 Check excess kurtosis: $K = \mathbb{E}[\varepsilon^4]/\sigma^4 - 3$

What Makes Financial Returns Special? IV

In practice, financial return series **almost always** reject H_0 of no ARCH effects. The question is not *whether* to model volatility but *which* GARCH specification to use.

Section 2

ARCH and GARCH Models

$$r_t = \mu_t + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \stackrel{\text{iid}}{\sim} (0, 1)$$

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \cdots + \alpha_q \varepsilon_{t-q}^2$$

Constraints: $\omega > 0$, $\alpha_i \geq 0$, $\sum \alpha_i < 1$ (stationarity).

$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2$$

Constraints: $\omega > 0$, $\alpha_i, \beta_j \geq 0$, $\sum \alpha_i + \sum \beta_j < 1$ (covariance stationarity).

In practice, GARCH(1,1) fits most financial series well:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

$\alpha + \beta$ close to 1: high persistence. Half-life = $\ln(0.5)/\ln(\alpha + \beta)$.

Typically: $\omega \approx$ small, $\alpha \approx 0.05$ – 0.15 , $\beta \approx 0.80$ – 0.90 .

Interpreting GARCH(1,1)

The GARCH(1,1) variance equation can be written as:

$$\sigma_t^2 = \frac{\omega}{1 - \alpha - \beta} + \alpha(\varepsilon_{t-1}^2 - \bar{\sigma}^2) + \beta(\sigma_{t-1}^2 - \bar{\sigma}^2)$$

where $\bar{\sigma}^2 = \omega / (1 - \alpha - \beta)$ is the **unconditional variance**.

Persistence and mean reversion

- $\alpha + \beta \rightarrow 1$: **IGARCH** (integrated GARCH) — variance shocks are permanent
- $\alpha + \beta < 1$: volatility mean-reverts to $\bar{\sigma}^2$
- Speed of mean reversion: $1 - \alpha - \beta$
- **Forecast**: $\sigma_{t+h|t}^2 \rightarrow \bar{\sigma}^2$ as $h \rightarrow \infty$

Multi-step volatility forecast

$$\sigma_{t+h|t}^2 = \bar{\sigma}^2 + (\alpha + \beta)^{h-1}(\sigma_{t+1|t}^2 - \bar{\sigma}^2)$$

Useful for option pricing (need vol at various horizons) and VaR over multiple holding periods.

Section 3

Asymmetric GARCH Models

Stylised fact: Negative return shocks increase volatility *more* than positive shocks of equal magnitude. Symmetric GARCH(1,1) misses this.

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 1_{(\varepsilon_{t-1} < 0)} + \beta \sigma_{t-1}^2$$

The **asymmetry parameter** $\gamma > 0$ means bad news ($\varepsilon_{t-1} < 0$) increases volatility by an extra $\gamma \varepsilon_{t-1}^2$.

$$\ln \sigma_t^2 = \omega + \beta \ln \sigma_{t-1}^2 + \alpha \left| \frac{\varepsilon_{t-1}}{\sigma_{t-1}} \right| + \gamma \frac{\varepsilon_{t-1}}{\sigma_{t-1}}$$

- Log formulation: $\sigma_t^2 > 0$ guaranteed without sign constraints
- $\gamma < 0$: leverage effect (negative standardised residuals increase log-variance)
- Asymmetry test: $H_0 : \gamma = 0$

Capturing the Leverage Effect III

Model selection in practice

Model	Captures	When to use
GARCH(1,1)	Clustering, persistence	Default starting point
GJR-GARCH	+ Leverage effect	Equities, FX
EGARCH	+ Leverage, no positivity constraint	High-kurtosis series
IGARCH	Near-unit-root persistence	Intraday, very long samples

Use AIC/BIC for model comparison. Also test residuals $z_t = \hat{\varepsilon}_t / \hat{\sigma}_t$:

- Ljung–Box on z_t : no serial correlation in mean
- Ljung–Box on z_t^2 : no remaining ARCH effects
- Excess kurtosis: choose t -distribution or GED if > 0 after GARCH

Section 4

Multivariate GARCH: DCC

Dynamic Conditional Correlation I

Motivation: Portfolio risk requires the **covariance matrix** Σ_t , not just individual volatilities. DCC (Engle, 2002) models this efficiently.

Step 1 — Univariate GARCH:

Estimate GARCH(p_i, q_i) for each return series i . Obtain standardised residuals $z_{i,t} = \varepsilon_{i,t} / \sigma_{i,t}$.

Step 2 — Dynamic correlation:

$$Q_t = (1 - a - b)\bar{Q} + az_{t-1}z'_{t-1} + bQ_{t-1}$$

$$R_t = \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2}$$

$$\Sigma_t = D_t R_t D_t, \quad D_t = \text{diag}(\sigma_{1,t}, \dots, \sigma_{n,t})$$

where $\bar{Q}$ is the unconditional correlation matrix; $a, b > 0$; $a + b < 1$.

Advantages of DCC

- Computationally feasible for large n (two-step estimation)
- Positive definite Σ_t by construction
- Time-varying correlations capture crisis periods (correlations spike during crises)
- Parsimonious: only 2 additional parameters (a, b) for correlation dynamics

Minimum variance portfolio weights:

$$w_t^* = \frac{\Sigma_t^{-1} \mathbf{1}}{\mathbf{1}' \Sigma_t^{-1} \mathbf{1}}$$

Using DCC-GARCH Σ_t gives **dynamic rebalancing**: portfolio weights change as volatilities and correlations evolve.

Christoffersen et al. (2012): international diversification benefits are time-varying and correlated with market stress.

Section 5

Value-at-Risk and Expected Shortfall

$\text{VaR}_\alpha(h)$ is the loss not exceeded with probability $1 - \alpha$ over a holding period of h days:

$$P(r_{t+1} < -\text{VaR}_\alpha) = \alpha$$

For GARCH(1,1) with Normal innovations:

$$\text{VaR}_\alpha = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \Phi^{-1}(\alpha)$$

where $\Phi^{-1}(\alpha)$ is the α -quantile of the standard normal.

Expected Shortfall (ES)

Also called CVaR. The expected loss **conditional on** the loss exceeding VaR:

$$ES_{\alpha} = \mathbb{E}[-r_{t+1} \mid r_{t+1} < -\text{VaR}_{\alpha}]$$

ES is a **coherent** risk measure; VaR is not (fails sub-additivity). Basel III/IV requires ES at 97.5%.

GARCH-based backtesting

Kupiec (1995): Are VaR violations at the stated frequency?

Christoffersen (1998): Are violations independent (no clustering)?

DQ test: Dynamic quantile test for GARCH-VaR.

If GARCH model is correct, violations should be Bernoulli iid with probability α .

Fat tails and distributional choice

Normal GARCH often underestimates tail risk. Better alternatives:

- **Student- t GARCH**: $z_t \sim t(\nu)$; ν estimated by MLE; heavier tails
- **Skewed- t (Hansen, 1994)**: allows asymmetry in distribution
- **GED (Generalised Error Distribution)**: $z_t \sim \text{GED}(\nu)$
- **Extreme Value Theory (EVT)**: model the tail of z_t with a Generalised Pareto Distribution (GPD) — most principled approach for stress testing

Financial regulation requires VaR/ES computation at 99% (Basel) or 99.5% (Solvency II) confidence levels. At these extreme quantiles, distributional choice is **critical**. GARCH + t or GARCH + EVT substantially outperform normal GARCH in backtests.

Core references

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Software: R (`rugarch`: ARCH/GARCH/EGARCH/GJR; `rmgarch`: DCC) or Python (`arch.arch_model`, `arch.multivariate`). Applied exercises: fit GARCH(1,1) and GJR-GARCH to IBOVESPA returns; compute DCC between IBOVESPA and S&P 500; backtest 99% VaR.

Next: Lecture 7

Forecasting and Model Selection

MSFE · Diebold–Mariano · GETS/Autometrics · LASSO · Forecast combination